HARSHITH KUMAR MANNEPALLI

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Professional Summary

Computer Science (AI & ML) student building full stack and AI-powered applications with Python, TypeScript, React, Next.js, FastAPI, and PostgreSQL. Experienced in containerized deployment with Docker, Kubernetes and CI/CD, with 5 pull requests merged into Cal.com and freeCodeCamp and two IEEE conference papers. Seeking an entry-level Software Engineer role with a focus on AI-driven products.

Education

Uttaranchal University — Dehradun, India
Bachelor of Technology in Computer Science (AI & Machine Learning) — *Expected May 2027*

Technical Skills

Languages: Python, TypeScript

Backend & Frontend: FastAPI, Node.js, REST APIs, React.js, Next.js

Databases: PostgreSQL, SQLite

DevOps & Cloud: Docker, Kubernetes, Git/GitHub, Microsoft Azure, Microsoft Foundry

Testing & Quality: Unit Testing, API Testing, Debugging, Code Review, Performance Optimization

AI/ML: LLM APIs, AI Agents, Model Evaluation, Claude code

Experience

Open Source Contributor — Cal.com & freeCodeCamp — Remote
calcom/cal.com | *freeCodeCamp/freeCodeCamp* — *March 2026 – Present*

- Got 5 pull requests merged into production codebases after maintainer code review, covering core library logic, CI/CD workflows, automated validation, and accessibility (WCAG) fixes.
- Cal.com (TypeScript/Node.js): Tracked down a Day.js UTC/timezone plugin load-order bug, shipped a targeted fix on a platform used by 50k+ teams, and verified it through cross-browser testing.
- freeCodeCamp: Built automated file-structure validation for workshops with directory scanning and unit tests; fixed a high-contrast navbar regression and added test coverage.

Projects

RecoveryOS | *TypeScript, Node.js, SQLite, REST APIs, Docker* | GitHub

- Designed an AI-driven revenue-recovery system where AI suggestions pass through a rule-based safety layer (risk limits, retry budgets, cooldowns, kill-switch) before any action is taken.
- Built idempotent webhook handling, an execution ledger with reconciliation, and an append-only audit trail; across 135 failed-payment test cases the AI lane recovered **20,726 extra** with **zero unsafe actions and zero duplicate executions**.
- Set up automated testing and release checks: **232/232 unit tests** and **36/36 validation checks** passing, with seeded simulations for reproducible results.

Smart Sales Intelligence Agent | *Python, FastAPI, Next.js, React* | GitHub

- Built a modular 4-stage AI workflow (Orchestrator → Research → Analysis → Outreach) with FastAPI and REST APIs, tracking latency, errors, and usage for each stage.
- Developed and deployed a Next.js 15 dashboard with real-time execution timelines so every workflow run can be traced and debugged.

Uni-Connect Hub | *React, TypeScript, Node.js, PostgreSQL* | GitHub | Live

- Built a student–alumni networking platform (selected in the Smart India Hackathon internal round) with role-based access for 200+ users across students, alumni, and admins.
- Tuned PostgreSQL joins and indexes to cut query time from 2.3s to 0.6s (**74% faster**).

Certifications

- **Oracle Cloud Infrastructure Generative AI Professional** — Oracle (2025) | ID: 322494826OCI25GAIOCP
- **AWS Solutions Architecture** — Amazon Web Services
- **Open Source Contribution** — GirlScript Summer of Code (GSSoC)

Research & Achievements

- **IEEE Research (June 2026):** Two conference papers: federated continual learning with EWC for IoT edge devices (93.1% accuracy, 38% less catastrophic forgetting than FedAvg) and AgriCogNet, an explainable ensemble for detecting cognitive fatigue in farm workers (91.3% accuracy, 0.962 AUC, under 38ms inference on Raspberry Pi 4).
- **Hackathons:** Smart India Hackathon internal round selection (Uni-Connect Hub); built and deployed two working web apps in 24 hours at the Dev Bhoomi University Lakshagriha Global Hackathon.